Chapter 3: AI as Offender Capability

Franklin Castillo

Abstract

Generative Artificial Intelligence (AI) has fundamentally altered the economics of institutional cybercrime, democratizing advanced offensive capabilities and dramatically lowering the cost of discovering and exploiting institutional vulnerabilities. Drawing upon routine activity theory and crime-and-place research, this chapter examines how AI acts as a potent force multiplier for threat actors, enabling them to systematically target institutions where weak guardianship meets attractive assets. By removing traditional barriers of time, language, and technical expertise, AI allows offenders to scale sophisticated attacks that disproportionately impact under-resourced colleges, hospitals, and local governments.

The chapter explores three primary vectors of AI-enabled exploitation: deepfake-enabled fraud that bypasses human verification protocols; AI-assisted reconnaissance and hyper-targeted phishing deployed at scale; and the automated generation of synthetic identities for "ghost student" enrollment fraud in higher education. Because these automated attacks exploit structural resource deficits, they amplify unequal risk across institutional sectors. To bridge theory and practice, the chapter concludes by introducing an AI Exposure Assessment framework designed for institutional risk registers. This practical tool provides under-resourced organizations with a structured methodology to evaluate their specific vulnerabilities to AI-augmented threats and strategically reinforce their guardianship without requiring enterprise-level security budgets.

Proposed Chapter Structure

1. Introduction: The New Economics of Cyber Offending

- **The Hook:** The shift from manual, labor-intensive cyber attacks to automated, AI-driven exploitation.
- **Theoretical Framing:** Contextualizing AI within the book's core framework. AI lowers the cost of identifying "attractive targets" and testing for "weak guardianship."
- **The Unequal Risk Paradigm:** Why institutions with limited security budgets (community colleges, municipal agencies, regional hospitals) are disproportionately vulnerable to this new scale of attack.

2. Vector I: Deepfake-Enabled Fraud and the Erosion of Trust

- **Mechanism of Attack:** How audio and video cloning technology (deepfakes) bypass traditional human-centric guardianship (e.g., voice verification for wire transfers).
- **Case Context:** Business Email Compromise (BEC) evolving into Business Identity Compromise.

- **Institutional Impact:** The severe financial and reputational consequences for under-resourced finance and HR departments lacking multi-person, cryptographic verification protocols.

3. Vector II: AI-Assisted Phishing and Automated Reconnaissance

- **Reconnaissance at Scale:** The use of Large Language Models (LLMs) to scrape institutional directories, academic publications, and public records to map organizational hierarchies automatically.
- **Hyper-Targeting:** Moving beyond generic "spray-and-pray" phishing. AI generating context-aware, grammatically perfect spear-phishing emails tailored to specific departments (e.g., grant administrators, procurement officers).
- **Bypassing Guardianship:** How these sophisticated lures evade standard, low-tier email security gateways.

4. Vector III: "Ghost Student" Enrollment Fraud

- **The Higher Education Vulnerability:** Exploiting the mandate for accessible admissions (an attractive target) against the reality of understaffed admissions and financial aid offices (weak guardianship).
- **AI's Role:** Utilizing generative AI to construct synthetic identities, generate unique application essays, and automate the completion of admissions forms and financial aid applications (FAFSA).
- **The Consequence:** Siphoning of federal and institutional financial aid, dilution of institutional metrics, and the massive administrative burden of unwinding synthetic enrollments.

5. Bridging Theory to Practice: The AI Exposure Assessment

- **Purpose:** Translating the theoretical threats into actionable intelligence for institutional leadership and Boards of Trustees.
- **Integration with Risk Registers:** How to formally categorize and weight AI-augmented threats alongside traditional cyber risks.
- **Assessment Categories:**
  - *Identity & Authorization Verification:* Auditing vulnerabilities to deepfakes.
  - *Public Data Exposure:* Assessing what internal hierarchy data is available for AI scraping.
  - *Process Automation Vulnerabilities:* Evaluating points in the student lifecycle (or patient intake) susceptible to synthetic flooding.
- **Mitigation for the Under-Resourced:** Low-cost, high-impact strategies (e.g., out-of-band verification, procedural friction) to improve guardianship.

6. Conclusion

- Summary of how AI shifts the balance of power toward the offender.
- Final reinforcement of the book's theme: While AI democratizes the *capability* to attack, the *impact* of those attacks remains deeply unequal, demanding targeted, proactive guardianship.